

VAISHNAVI SAPKALE

+91 9589806986 ◇ Pune, Maharashtra

[Email](#) ◇ [LinkedIn](#) ◇ [GitHub](#) ◇ [X](#)

SUMMARY

Software Engineer with 1+ year of experience building **scalable web applications**, **real-time systems**, and **AI-powered developer tools**. Strong in TypeScript, Node.js, React, PostgreSQL, Redis, and WebSockets, with hands-on experience in **distributed architectures**, custom matching engines, and **LLM tool-calling agents**. Focused on backend engineering, distributed systems, and developer infrastructure.

EXPERIENCE

Junior Software Developer

Feb 2024 - May 2025

Payroll: Harjai Computers Pvt. Ltd.

(Client: Bajaj Allianz General Insurance Company)

Pune, Maharashtra

- Built **reusable configurable components** for Mystique, an internal low-code platform enabling rapid application development across multiple business teams.
- Developed **secure enterprise web applications** by building frontend modules and backend REST APIs with JWT authentication and Role-Based Access Control (RBAC).
- Collaborated with cross-functional frontend and backend teams to deliver production features while improving maintainability and performance.

SKILLS

Languages: TypeScript, JavaScript, Java

Backend: Node.js, Express.js, Bun, REST APIs, WebSockets

Databases: PostgreSQL, Redis, MongoDB, Prisma ORM

AI Engineering: Gemini API, LLM Tool Calling, Agentic Workflows, Prompt Engineering

Frontend: React.js, Angular, Tailwind CSS

Tools: Git, GitHub, Docker, Postman

PROJECTS

HarnessCLI – Autonomous AI Coding Agent

[GitHub](#) — [Live](#) — [npm](#)

Tech Stack: TypeScript, Bun, Gemini API, React (Ink)

- Built an **autonomous AI coding agent** that understands natural-language instructions to analyze repositories, modify source code, execute terminal commands, and iteratively solve development tasks.
- Designed a **provider-agnostic LLM abstraction layer** enabling seamless integration across multiple AI providers through a unified interface.
- Published **harness-terminal** as an npm package and engineered a modular agent runtime featuring persistent memory, tool-calling orchestration, automatic retries, safety guardrails, and an interactive CLI/TUI.

Centralized Exchange (CEX) — Matching Engine Simulation

[GitHub](#) — [Live](#)

Tech Stack: TypeScript, Express, PostgreSQL, Redis, Prisma, WebSockets, React

- Built a **simulated centralized exchange** to understand how order matching, trade execution, and exchange architectures work.
- Implemented a **price-time priority matching engine** supporting market and limit orders, maintaining an in-memory order book for simulated trades.
- Designed a **distributed architecture** separating the API server and matching engine using Redis Pub/Sub, with PostgreSQL for persistence and WebSockets for real-time order book updates.

EDUCATION

Bachelor of Technology in Computer Science, RGPV University

June 2023

CGPA: 8.4/10